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Significantly enhanced electrical properties in $\text{CaBi}_2\text{Nb}_2\text{O}_9$ -based high-temperature piezoelectric ceramics ✓Changbai Long ✉ ; Bo Wang; Wei Ren ✉; Kun Zheng; Huiqing Fan ; Dawei Wang ; Laijun Liu ✉ *Appl. Phys. Lett.* 117, 032902 (2020)<https://doi.org/10.1063/5.0007189>**Articles You May Be Interested In**Ferroelectric properties of lead-free polycrystalline $\text{CaBi}_2\text{Nb}_2\text{O}_9$ thin films on glass substrates*AIP Advances* (March 2016)Low-temperature preparation and electrical properties of $\text{CaBi}_2\text{Nb}_2\text{O}_9$ piezoelectric ceramic by addition of B_2O_3 *J. Appl. Phys.* (September 2022)High Curie point $\text{Ca Bi}_2 \text{Nb}_2 \text{O}_9$ thin films: A potential candidate for lead-free thin-film piezoelectrics*J. Appl. Phys.* (October 2006)**Applied Physics Letters****Special Topics Open
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Significantly enhanced electrical properties in $\text{CaBi}_2\text{Nb}_2\text{O}_9$ -based high-temperature piezoelectric ceramics

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ABSTRACT

A synergetic effect of Li/Nd at the A site and W/Mn at the B site co-substitution on electrical properties of calcium bismuth niobate ($\text{CaBi}_2\text{Nb}_2\text{O}_9$, CBNO)-based ceramics is proposed in this paper. CBNO primarily shows mixed p -type and ionic conduction due to a trace of oxygen vacancies ($\text{V}_{\text{O}}^{\bullet\bullet}$). Both bulk resistivity (ρ_b) and activity energy (E_a) for conduction are dramatically increased by the co-substitution in $\text{Ca}_{0.92}(\text{Li}_{1/2}\text{Nd}_{1/2})_{0.08}\text{Bi}_2\text{Nb}_{2-x}(\text{W}_{3/4}\text{Mn}_{1/4})_x\text{O}_9$ [CBNO-LN- x (WM), $x = 0-0.02$]. Consequently, the modified compositions almost show an intrinsic conducting mechanism due to a significant suppression of the $\text{V}_{\text{O}}^{\bullet\bullet}$ concentration. Furthermore, the synergetic effect of Li/Nd at the A site and W/Mn at the B site greatly improves the piezoelectricity coefficient (d_{33}), which is up to ~ 21.1 pC/N in the optimized composition [CBNO-LN-10(WM)]. This combined with a high Curie temperature ($T_C = \sim 946^\circ\text{C}$) and high resistivity indicates that CBNO-LN-10(WM) is a good promising material in high-temperature piezoelectric applications.

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Piezoelectric materials with high Curie temperature (T_C) have received increasing attention due to their potential applications in the fields of aerospace, automotive, power plants, and material processing.^{1,2} To date, single crystal of quartz (SiO_2) is still the main commercially high-temperature piezoelectric material. The shortcomings of quartz are relatively low piezoelectric coefficient ($d_{33} \sim 2$ pC/N) and low working temperature ($\leq 350^\circ\text{C}$).¹⁻³ Other single crystals such as LiNbO_3 , $\text{YCa}_4\text{O}(\text{BO}_3)_3$, $\text{GdCa}_4\text{O}(\text{BO}_3)_3$, $\text{Sr}_2\text{Nb}_2\text{O}_7$, and $\text{La}_2\text{Ti}_2\text{O}_7$ show ultrahigh T_C ($\sim 1000-1400^\circ\text{C}$) and comparable d_{33} ($\sim 4-7$ pC/N).³⁻⁶

High T_C ferroelectric ceramics have advantages over the above single crystals, including lower cost and the ease of tailoring their properties by changing the chemical composition and preparation/heat-treatment processing.² Aurivillius-phase ferroelectric ceramics with high T_C ($\geq 650^\circ\text{C}$) are good candidate materials in high-temperature

piezoelectric applications. Among them, calcium bismuth niobate ($\text{CaBi}_2\text{Nb}_2\text{O}_9$, CBNO) ceramic has the highest T_C ($\sim 940^\circ\text{C}$), as well as the highest thermal-depoling temperature (up to 800°C).^{7,8} Therefore, it is expected to be applied to the working conditions at high- and ultrahigh-temperatures. Although the spontaneous polarization (P_S) of CBNO reaches $\sim 24.4 \mu\text{C}/\text{cm}^2$, the d_{33} of its ordinarily fired (OF) ceramics is only $\sim 5-7.5$ pC/N.⁷⁻⁹

Much work has been focused on improving d_{33} of CBNO ceramics, which is elevated to $\sim 12.6-17.5$ pC/N via appropriate substitutions at A and/or B sites.⁹⁻¹⁵ Among those substitutions, LiLn (Ln = La, Ce, Nd, Sm, Pr, etc., or their mixture) isovalent substitutions at the A site are very impressive.¹³⁻¹⁵ Enhanced piezoelectric properties induced by LiLn substitutions are also reported for other Aurivillius-phase ferroelectric ceramics, e.g., LiCe substituted $\text{CaBi}_4\text{Ti}_4\text{O}_{15}$ and

$\text{Na}_{0.5}\text{Bi}_{2.5}\text{Nb}_2\text{O}_9$.^{16,17} In the study of perovskite piezoelectric ceramics, it is found that multi-valent Mn substitution is an effective way to improve the electrical properties, e.g., Mn substituted $\text{Bi}_{0.5}\text{Nb}_{0.5}\text{TiO}_3$ and $\text{K}_{0.5}\text{Na}_{0.5}\text{NbO}_3$.^{18–21} In this work, we developed a synergetic effect to increase piezoelectricity, ferroelectric polarization, and electrical resistivity of CBNO ceramics by the co-substitution of $(\text{Li}_{1/2}\text{Nd}_{1/2})$ at the A site and $(\text{W}_{3/4}\text{Mn}_{1/4})$ and B site. The collocation of $\frac{3}{4}\text{W}^{6+}$ and $\frac{1}{4}\text{Mn}^{2+}$ was selected to balance the charge valence of the B site.

Polycrystalline ceramics of CBNO and modified $\text{Ca}_{0.92}(\text{Li}_{1/2}\text{Nd}_{1/2})_{0.08}\text{Bi}_2\text{Nb}_{2-x}(\text{W}_{3/4}\text{Mn}_{1/4})_x\text{O}_9$ [CBNO-LN- x (WM), $x = 0, 0.005, 0.01, 0.015$, and 0.02] were prepared by a traditional solid-state reaction. The starting raw materials [Bi_2O_3 (99.5%), Nb_2O_5 (99.9%), CaCO_3 (99.8%), Li_2CO_3 (99.9%), Nd_2O_3 (99.99%), WO_3 (99.9%), MnO (99.99%); Sinopharm Chemical Reagent Co., Ltd, CN] were mixed by ball milling, dried, and then calcined at 850°C for 4 h. Dense ceramics were prepared by sintering at 1140 – 1160°C for 2 h. Dielectric properties were measured by using an LCR meter (E4980A, Agilent, CA, USA). Ferroelectric polarization-, field-induced strain-, and current density-electric field (P - E , S - E , and J - E , respectively) hysteresis loops were measured by using a ferroelectric analyzer (TF-3000, Aix ACCT, Aachen, Germany). Electron paramagnetic resonance (EPR) measurements were performed on a high sensitivity X-band (~ 9.2 GHz) EPR system (JES-FA200, JEOL, Japan). Domain structures were characterized by using a commercial piezoelectric force microscope (PFM; MFP-3D, Asylum Research, Goleta, CA). After being poled in silicone oil under a dc electric field of 160 – 270 kV/cm for 20 – 40 min at 160°C , d_{33} of the specimens was determined by using a piezo- d_{33} meter (ZJ-6, Institute of Acoustics, Chinese Academic of Science, Beijing, China) at room temperature. Impedance spectra (IS) were performed by using a Solartron impedance analyzer (SI 1260, Hampshire, U.K.). Electrodes were made of fired-on platinum pastes at 900°C for 30 min.

On inspecting the XRD patterns of the modified CBNO-LN- x (WM), small amounts of impurity phases ($\text{Bi}_2\text{W}_2\text{O}_9$) are detected in the $x = 0.02$ composition (Fig. S1, supplementary material). Evolution in unit cell volume (V) extracted from the XRD results reveals a shrinkage of V for CBNO-LN-0(WM) compared with that for CBNO (Fig. S2). This should be attributed to the fact that the average size of $(\text{Li}^{+}_{1/2}\text{Nd}^{3+}_{1/2})$ (1.21 Å, 12 CN) is smaller than that of Ca^{2+} (1.34 Å; 12 CN).²² Similarly, the replacement of Nb^{5+} (0.64 Å; 6 CN) by the smaller size $(\text{W}_{3/4}\text{Mn}_{1/4})$ ($\text{W}^{6+}_{3/4}\text{Mn}^{2+}_{1/4}$: ~ 0.63 Å, $\text{W}^{6+}_{3/4}\text{Mn}^{4+}_{1/4}$: ~ 0.58 Å; 6 CN) results in V decreasing continuously.²² Viewed from cross sectional SEM images, CBNO and CBNO-LN-0(WM) show a comparable average grain size (length) (~ 6.1 – 6.5 μm , Fig. S3). In contrast, the size of CBNO-LN-10(WM) and CBNO-LN-20(WM) increases dramatically to ~ 11.2 and ~ 13.5 μm , respectively. The possible reason for the rapid grain growth could be associated with small amounts of unreacted powders as sintering aids. However, they cannot be detected by XRD due to the determination limit of the instrument.

The crystal structure of selected CBNO-LN-0(WM) and CBNO-LN-10(WM) was identified by the Rietveld method using the GSAS-EXPGUI program [Figs. 1(a) and 1(b)]. Here, the coordinates of CBNO (orthorhombic, space group: $A2_1am$) reported by Blake *et al.* were used as a structure model.²³ For each composition, the refined profile fits well to the corresponding XRD pattern (Fig. S4). With the aim of reducing the overbonding of the A site cations,²³ low

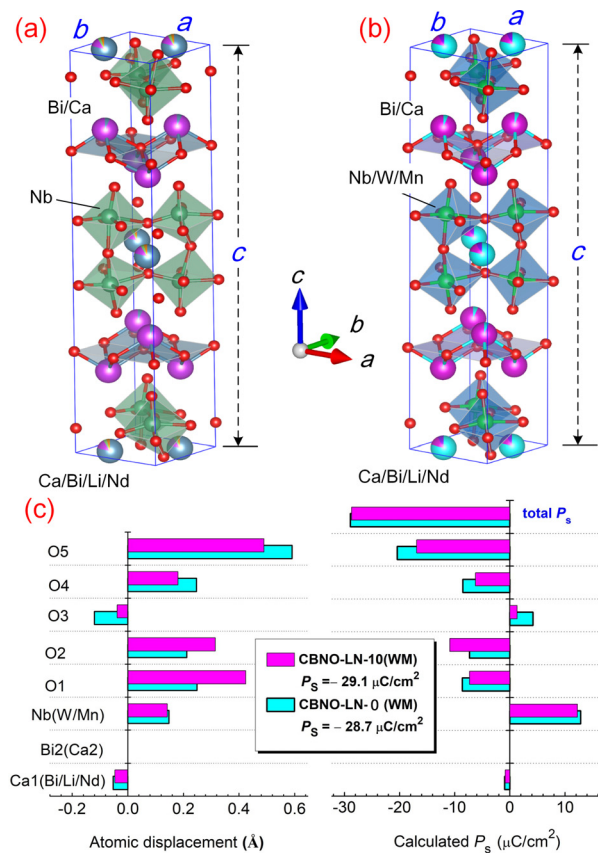


FIG. 1. Crystal structure of (a) CBNO-LN-0(WM) and (b) CBNO-LN-10(WM) at room temperature. (c) Displacement and contribution to total P_s of each constituent atom in CBNO-LN-0(WM) and CBNO-LN-10(WM).

disordering between the perovskite A-site cation and the bismuth cation in the $(\text{Bi}_2\text{O}_2)^{2+}$ layer is shown; i.e., there is ~ 6.3 – 6.4% Ca at the bismuth site of the $(\text{Bi}_2\text{O}_2)^{2+}$ layer, while there is ~ 12.6 – 12.8% Bi at the perovskite A site. The two fractions are higher than those of CBNO ($\sim 5.3\%$ and $\sim 10.5\%$, respectively).²³ It suggests that the substitutions of LiNd and WMn give rise to local lattice distortion in CBNO, which is in favor of an enhancement in the overbonding of the A-site cations. Furthermore, P_s of CBNO-LN-0(WM) and CBNO-LN-10(WM) can be calculated by using Shimakawa's mode²⁴

$$P_s = \sum_i \frac{m_i \times \Delta x_i \times Q_i e}{V}, \quad (1)$$

where m_i is the site multiplicity, Δx_i is the atomic displacement along the a axis from the corresponding position in the parent structure, and $Q_i e$ is the ionic charge for the i th constituent ion. Based on the atomic parameters (Tables S1 and S2), P_s of CBNO-LN-0(WM) and CBNO-LN-10(WM) was calculated to be ~ 28.7 and 29.1 $\mu\text{C}/\text{cm}^2$ [Fig. 1(c)], respectively, which are both higher than that of CBNO (~ 24.4 $\mu\text{C}/\text{cm}^2$).⁸

Temperature-dependent dielectric permittivity (ϵ') indicates that T_C of CBNO and CBNO-LN-0(WM) is ~ 940 and $\sim 947^\circ\text{C}$, respectively [Fig. 2(a)]. T_C of CBNO-LN-5(WM) is elevated to $\sim 950^\circ\text{C}$ and

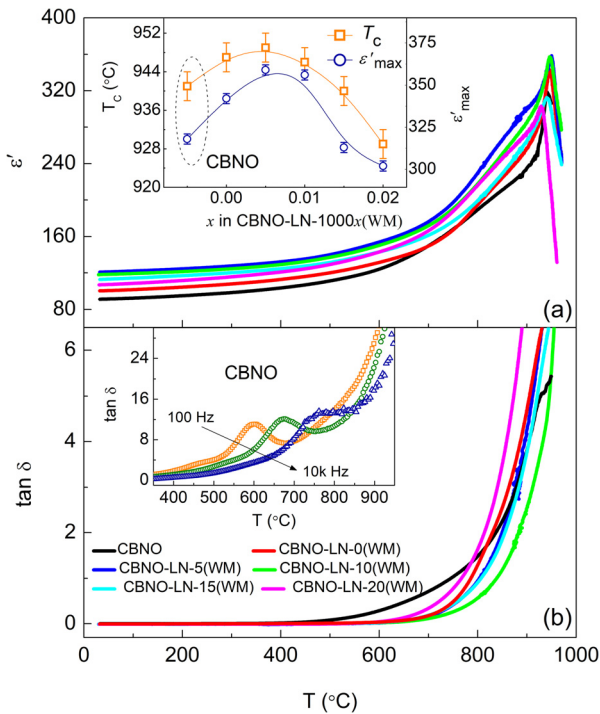


FIG. 2. Temperature dependences of (a) dielectric permittivity (ϵ') and (b) loss tangent ($\tan \delta$) at 1M Hz for all compositions. In (a), the inset shows the extracted T_C and $\epsilon'_{\max}$ (the maximum ϵ' at T_C). In (b), the inset shows the $\tan \delta$ vs T plots of CBNO at 0.1k, 1k, and 10k Hz.

then it decreases gradually to $\sim 929^\circ\text{C}$ with further increase in the x value in CBNO-LN- x (WM). The change of $\epsilon'_{\max}$ (maximum ϵ' at T_C) is similar to that of T_C . In addition, the substitutions of LiNd and WMn depress the loss tangent ($\tan \delta$) of CBNO (especially in the range of 400 – 800°C) [Fig. 2(b)]. For CBNO, distinct loss peaks are observed between 500 and 850°C at lower frequencies (0.1 – 10 kHz) [the inset of Fig. 2(b)]. However, this characteristic is absent in the modified CBNO-LN- x (WM) (not shown here). The position of the loss peak moves toward higher temperature as the probing frequency increases, which is typical of a thermally activated Debye-like relaxation.²⁵ $\text{Bi}_4\text{Ti}_3\text{O}_{12}$, acceptor $\text{Bi}_3\text{TiNbO}_3$, and CBNO ceramics showed similar behaviors, and the authors suggested that the relaxation was mainly attributed to an oxygen-ion-jump loss mechanism.^{10,25,26}

The influence of LiNd and WMn co-substitutions on ferroelectric properties and field-induced strain is displayed in Figs. S5 and S6, respectively. It is clear that the modified compositions show saturated P - E loops and well-shaped “butterfly” S - E loops. The extracted remnant polarization (P_r) and the field-induced strain (S_{33}) of CBNO and CBNO-LN- x (WM) are given in Fig. 3(a). P_r and S_{33} of CBNO-LN- x (WM) are ~ 7.0 – $10.9 \mu\text{C}/\text{cm}^2$ and ~ 4.3 – 7.6×10^{-4} , respectively, which are much higher than those of CBNO ($\sim 3.4 \mu\text{C}/\text{cm}^2$ and $\sim 2.4 \times 10^{-4}$, respectively). For CBNO-LN-0(WM) and CBNO-LN-10(WM), the P_r (~ 7.0 and $10.9 \mu\text{C}/\text{cm}^2$, respectively) is much lower than the calculated P_s . Since the OF Aurivillius-phase ceramics have the characteristic of randomly oriented ceramic grains, the obtained P_r is a compromise value in polarizations of different crystallographic directions [i.e., $a(b)$ axis and c axis].

Coercive field (E_c) is indicated by the site of current peaks in the J - E loops. The E_c is estimated to be $\sim 160 \text{ kV}/\text{cm}$ for CBNO and increases to ~ 168 – $188 \text{ kV}/\text{cm}$ for CBNO-LN- x (WM) [Fig. 3(a)]. The A-site cation vacancies in the perovskite structures can make the

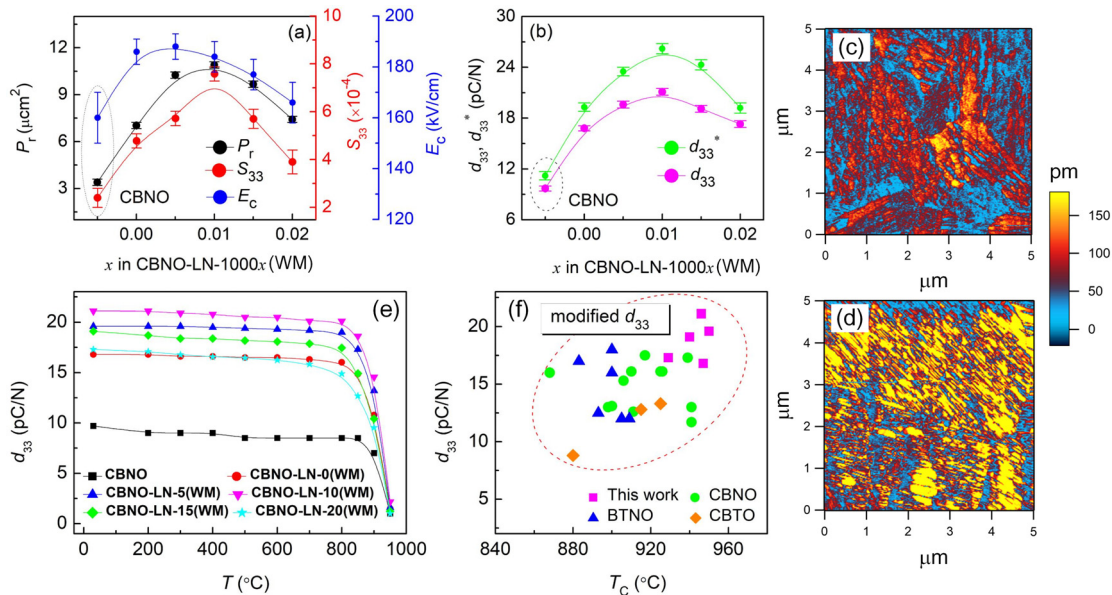


FIG. 3. (a) Extracted P_r , S_{33} , E_c and (b) d_{33}^* , d_{33} of all compositions. Vertical PFM (V-PFM) images of (c) CBNO and (d) CBNO-LN-10(WM). (e) Effects of annealing temperature on d_{33} for all compositions. (f) A comparison in d_{33} between the modified CBNO-LN- x (WM) and other reported Aurivillius-phase piezoelectric [CBNO, $\text{Bi}_3\text{TiNbO}_3$ (BTNO), $\text{CaBi}_2\text{Ta}_2\text{O}_9$ (CBTO)] ceramics with ultrahigh T_C ($\geq 850^\circ\text{C}$).

movement of domain walls easier.²⁷ For Aurivillius-phase ferroelectrics, it would be a similar effect if the cation vacancies (i.e., bismuth vacancies, V_{Bi}''') are present in the $(\text{Bi}_2\text{O}_2)^{2+}$ layers.²⁵ In the case of CBNO, the introduction of V_{Bi}''' could allow the ease of polarization through the existing oxygen vacancies (V_{O}''), which interact with the domain walls. Lowering the V_{Bi}''' concentration, as well as the loss of Bi_2O_3 , may lead to an increase in E_c for the modified CBNO-LN- x (WM) (see Fig. S9, discussed later). In addition, the charged substitution cations (e.g., Li_{Ca}' , Nd_{Ca}^* , and W_{Nb}^*) may have an influence on their ferroelectric polarization.

In Fig. 3(b), the d_{33} of CBNO-LN- x (WM) reaches up to ~ 16.8 – 21.1 pC/N, which is approximately twice higher than that of CBNO. Furthermore, the modified compositions present a higher reverse piezoelectric coefficient ($d_{33}^* = 19.2$ – 26.2 pC/N), which can be calculated from the slope of S – E curves at 0 kV/cm. The PFM measurements also indicate that the piezoresponse of the modified CBNO-LN- x (WM) is stronger than that of pure CBNO. In Figs. 3(c) and 3(d), CBNO-LN-10(WM) shows a larger vertical amplitude with the same stimulating electric field (1 V) compared with CBNO. Figure 3(e) demonstrates the effect of thermal depoling on d_{33} of all compositions. Here, the remaining d_{33} was measured at room temperature after annealing for 2 h at the depoling temperature on the x -axis. The d_{33} of the specimens decreases by ~ 5 – 13% after annealing at 800°C . It suggests that they demonstrate good piezoelectric thermal stability. Piezoelectricity of the specimens quickly disappears as the temperature further increases to approximation T_C .

As discussed in Refs. 17 and 28, the piezoelectricity of ferroelectric ceramics primarily originates from the intrinsic lattice contribution and the extrinsic contribution due to the movements of non- 180° domain walls. In the modified CBNO-LN- x (WM), the improvement of d_{33} and P_r should be compatible. First, an increase in P_s is induced by LiNd and WMn co-substitutions. For example, the P_s of CBNO-LN-10(WM) is ~ 29.1 $\mu\text{C}/\text{cm}^2$, which is higher than that of CBNO (~ 24.4 $\mu\text{C}/\text{cm}^2$). Second, a higher external field is applied to the poling process, which is helpful for the specimens achieving higher P_r and d_{33} . This is owing to the significant increase in the electrical resistivity of CBNO-LN- x (WM) (discussed later). Finally, the coarse grains lead to the release of internal stress due to the decrease in the volume fraction of grain boundaries. The release of stress is beneficial to the switching of domains along the external field during poling.²⁵ As a result, the d_{33} of the modified CBNO-LN- x (WM) (~ 16.8 – 21.1 pC/N) is extremely well comparable to or higher than that of the reported ultrahigh- T_C ($\geq 850^\circ\text{C}$) Aurivillius-phase ceramics^{9–15,25,29–32} [Fig. 3(f)].

As ferroelectric materials, high resistivity (ρ) is very important for their high-temperature piezoelectric applications. Also, high ρ ensures them to be sufficiently polarized under a large external electric field for obtaining higher P_r and d_{33} .⁶ Here, the impedance spectra of CBNO and CBNO-LN- x (WM) were measured in the temperature range of 400 – 800°C . It reveals that modified CBNO-LN- x (WM) ceramics contain one electrical component in essence at 650°C , i.e., only the bulk effect [Figs. 4(a)–4(c)]. For CBNO, the low frequency data presenting a small additional arc associates with the electrode polarization (Warburg impedance). The weak ionic conduction is

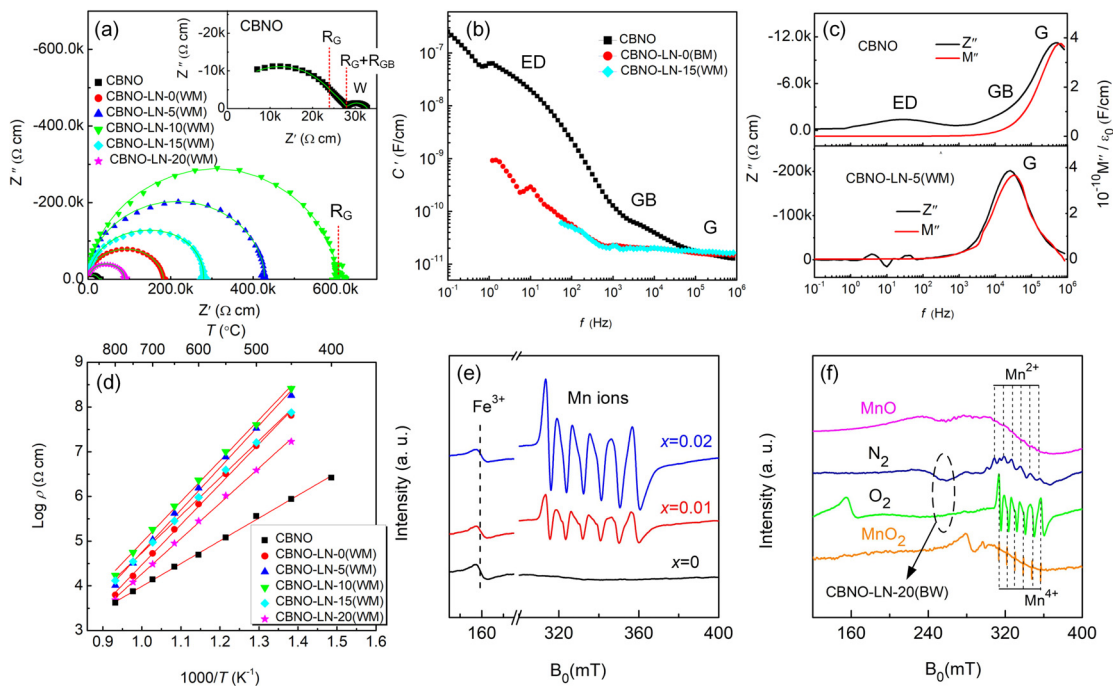
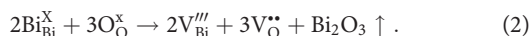


FIG. 4. (a) Complex Z^* plots of CBNO (inset) and modified CBNO-LN- x (WM) at 650°C . (b) Capacitance (C') spectroscopic plots of CBNO, CBNO-LN-0(WM), and CBNO-LN-15(WM) at 650°C . (c) Z''/M'' spectroscopic plots of CBNO and CBNO-LN-5(WM) at 650°C . (d) Arrhenius plots of ρ_b for all compositions at 400 – 800°C . (e) EPR spectra of CBNO, CBNO-LN-10(WM), and CBNO-LN-20(WM) at 100 K. (f) EPR spectra of CBNO-LN-20(WM) at 100 K after thermal annealing under N_2 and O_2 , and the EPR spectra of MnO and MnO_2 powders are also given for comparison.

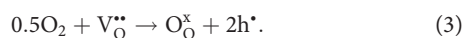
induced by the migration of $V_{O}^{\bullet\bullet}$. Furthermore, the impedance spectra of selected CBNO, CBNO-LN-0(WM), and CBNO-LN-10(WM) were measured under different atmospheres (air, N_2 , and O_2). It can be found the CBNO is primarily a p -type (h^*) conductor with a small amount of oxygen ionic conduction at 700 °C (Fig. S7). In contrast, the intrinsic electronic conduction (n type) dominates the conduction mechanism of both CBNO-LN-0(WM) and CBNO-LN-10(WM).

Figure 4(d) shows the Arrhenius plots of bulk resistivity (ρ_b) for all compositions. It is clear that the ρ_b of CBNO dramatically increases by introducing LiNd and WMn co-substitutions. Among them, CBNO-LN-10(WM) shows the highest ρ_b ($\sim 2 \times 10^6$ and $\sim 2 \times 10^5 \Omega \text{ cm}$ at 600 and 700 °C, respectively). In addition, the modified CBNO-LN- x (WM) exhibits a high thermal activity energy (E_a) of ~ 1.56 – 1.87 eV . For CBNO-LN-0(WM) and CBNO-LN-10(WM), the E_a is close to half of the calculated bandgap (E_g) (3.58 and 3.94 eV, respectively; Fig. S8), which indicates an intrinsic thermal activated conduction in the temperature range (450–800 °C).^{13,14,33} However, the E_a ($\sim 1.01 \text{ eV}$) of CBNO is much lower than half of its calculated E_g (3.70 eV; Fig. S8). This is relative to the mixed p -type (h^*) and ionic conduction mechanism in it. The barrier energy for the $V_{O}^{\bullet\bullet}$ conducting was reported to be only ~ 0.4 – 0.5 eV for $\text{Bi}_4\text{Ti}_3\text{O}_{12}$ - and $\text{Na}_{0.5}\text{Bi}_{0.5}\text{TiO}_3$ -based ceramics.^{34–36}

In CBNO, the generation of $V_{O}^{\bullet\bullet}$ primarily associates with the volatilization of Bi_2O_3 during sintering



The O_2 atmosphere would encourage the absorption of oxygen on the specimen surface, which is described as

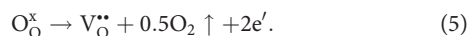


According to Eqs. (2) and (3), CBNO shows mixed p -type (h^*) and ionic conduction. Viewed from Fig. S9, the loss of Bi_2O_3 in CBNO-LN-0(WM) after sintering at 1150 °C is only $\sim 0.71 \text{ at. \%}$, which is much lower than that in CBNO ($\sim 1.37 \text{ at. \%}$). This implies that the $\text{Li}^+/\text{Nd}^{3+}$ isovalent substitution effectively suppresses the $\text{V}_{\text{Bi}}^{\text{X}}$ and $\text{V}_{\text{O}}^{\bullet\bullet}$ concentrations. Moreover, the two concentrations are possibly very low in the perovskite layers, which predominate in the electrical conductivity of the system because the Bi ions mainly locate at the site of the insulating $(\text{Bi}_2\text{O}_2)^{2+}$ layer. Consequently, CBNO-LN-0(WM) shows higher ρ_b and E_a compared with CBNO.

When WMn substitution appears at the B site, the ionization of the initial Mn^{2+} will further decrease the $\text{V}_{\text{O}}^{\bullet\bullet}$ concentration in CBNO-LN- x (WM) ($x = 0.005$ – 0.02) ceramics,



So, the introduction of Mn can be considered as donor substitution. However, a high level of WMn substitution may lead to excess electrons, which is described as $\text{Mn}^{2+} \rightarrow \text{Mn}^{4+} + 2e^-$. As a consequence, ρ_b first increases and then decreases as the x value in CBNO-LN- x (WM) increases from 0 to 0.02. For CBNO-LN-0(WM) and CBNO-LN-10(WM) [Figs. S7(b) and S7(c)], the required electrons for decreasing ρ_b in a reducing atmosphere (N_2) are probably created through the oxygen loss reaction,



The coexistence of Mn^{2+} and Mn^{4+} can be verified by EPR results, as shown in Fig. 4(e). In the EPR spectra of CBNO-LN-10(WM) and

CBNO-LN-20(WM), signals associated with multi-valent Mn ions (Mn^{2+} , Mn^{4+}) are observed in the magnetic field of ~ 310 – 360 mT . This is consistent with the results of Mn doped $\text{K}_{0.5}\text{Na}_{0.5}\text{NbO}_3$ and PbTiO_3 .^{18,37,38} It is interesting that the EPR spectrum of CBNO-LN-20(WM) after annealing under N_2 at 850 °C is similar to that of MnO [Fig. 4(f)], and a distinct Mn^{2+} -sextet hyperfine pattern is observed ($g \sim 2.003$). It suggests that Mn^{4+} is reduced to Mn^{2+} . In contrast, the spectrum presents no obvious change and still shows coexisting Mn^{2+} and Mn^{4+} after annealing under O_2 . By means of the spectrum of MnO_2 , the signal of Mn^{4+} is characterized by the g factor of ~ 1.997 . In addition, small amounts of impurity Fe^{3+} ($g = 4.27$) can be found in the EPR measurement, which is abundant in nature [Fig. 4(e)].^{39,40}

In summary, the structure and electrical properties of CBNO-based ceramics were tailored by Li/Nd at the A site and W/Mn at the B site co-substitution. Modified compositions [CBNO-LN- x (WM)] show significant enhancements in d_{33} , d_{33}^* , P_r , and S_{33} , which are $\sim 21.1 \text{ pC/N}$, $\sim 26.2 \text{ pC/N}$, $\sim 10.9 \mu\text{C/cm}^2$, and $\sim 7.6 \times 10^{-4}$, respectively, obtained in the $x = 0.01$ composition. In addition, the conducting mechanism changes from the mixed p -type and ionic conduction (CBNO) to the intrinsic conduction (n -type) due to a significant suppression in the $\text{V}_{\text{O}}^{\bullet\bullet}$ concentration induced by the co-substitution. The study supports an effective way to adjust the chemical composition in CBNO by the ionic synergetic effect for high-temperature piezoelectric applications.

See the [supplementary material](#) for XRD patterns, SEM images, lattice parameters, Rietveld refinements, P - E loops, S - E curves, variable-atmosphere impedance data, band structures, and loss of Bi_2O_3 .

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DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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